

**“MORPHOLOGICAL VARIABILITY IN
RHIZOCTONIA BATATICOLA (TAUB.) BUTLER
ISOLATES CAUSING DRY ROOT ROT OF
CHICKPEA IN WESTERN MAHARASHTRA”**

By

Mr. Sachin Subhash Doltade

(Reg. No. R/013/136)

A thesis submitted to the

**MAHATMA PHULE KRISHI VIDYAPEETH,
RAHURI - 413 722, DIST. AHMEDNAGAR,
MAHARASHTRA, INDIA**

in partial fulfilment of the requirements for the degree

of

MASTER OF SCIENCE (AGRICULTURE)
in

PLANT PATHOLOGY

**DEPARTMENT OF PLANT PATHOLOGY AND
AGRICULTURAL MICROBIOLOGY**

**POST GRADUATE INSTITUTE
MAHATMA PHULE KRISHI VIDYAPEETH,
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MAHARASHTRA STATE (INDIA)**

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2015

CANDIDATE'S DECLARATION

***I hereby declare that this thesis or part
there of has not been submitted
by me or other person to any
other University or Institute
for a Degree or
Diploma***

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C E R T I F I C A T E

This is to certify that the thesis entitled,
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SCIENCE (AGRICULTURE)** in **PLANT PATHOLOGY**, is a record
of *bona fide* research work carried out by **Mr. SACHIN
SUBHASH DOLTADE**, under my guidance and supervision and
that no part of the thesis has been submitted for any other
degree or diploma.

The assistance and help received during the course of
this investigation and sources of literature referred to have been
duly acknowledged.

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Mr. SACHIN SUBHASH DOLTADE, under the guidance and
supervision of **Dr. S.B. LATAKE**, Assistant Plant Pathologist,
Cotton Improvement Project, MPKV, Rahuri and that no part of
the thesis has been submitted for any other university for degree
or diploma.

Place: MPKV, Rahuri
Dated: / /2015.

(B. R. Ulmek)

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LIST OF ABBREVIATIONS

%	:	Per cent
µm	:	Micrometer
/	:	Per
@	:	At the rate of
BOD	:	Biological Oxygen Demand
°C	:	Degree Celsius
C.D.	:	Critical Difference
cm	:	Centimeter
cv.	:	Cultivar
<i>et al.</i>	:	Et alia
etc.	:	Et cetra
Fig	:	Figure
f. sp	:	Forma specialis
g	:	Gram
ha	:	Hectare
hrs.	:	Hours
i.e.	:	Id est (That is)
kg	:	Kilogram (s)
lb	:	Pound s
lit.	:	Liter
mm	:	Millimeter
ml	:	Milliliter
No.	:	Number
PDA	:	Potato dextrose agar
S.E.	:	Standard error
spp.	:	Species
<i>viz.</i>	:	Videlicet (Namely)

ABSTRACT

MORPHOLOGICAL VARIABILITY IN *RHIZOCTONIA* *BATATICOLOA* (TAUB.) BUTLER ISOLATES CAUSING DRY ROOT ROT OF CHICKPEA IN WESTERN MAHARASHTRA

By

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MASTER OF SCIENCE (AGRICULTURE)

in

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Research Guide : Dr. S.B. Latake

Department : Plant Pathology & Agricultural Microbiology

The present study was conducted to determine morphological, cultural and pathogenic variability amongst twelve isolates of *Rhizoctonia bataticola* collected from different locations of Western Maharashtra.

On the basis of colony growth, eight isolates were found to be fast growing (more than 80 mm), two moderately growing (between 70-80 mm) and remaining two were slow growing (less than 70 mm) on PDA. Four isolates showed compact and dense and two less dense colony (mycelium growth). Four isolates expressed whitish grey to smoky grey colour and four isolates showed grey to black colour. However, remaining isolates were having blackish mycelial colour. The size of sclerotia in isolates ranged from 52.3×54.39 to 104.06×113.52 µm. In case of sclerotial shape, five isolates were having oval shaped and remaining had round shaped sclerotia. According to days required to form sclerotia, early sclerotial formation was recorded in six isolates while in remaining isolates late sclerotial formation was observed. Isolate RB-10 (Jalgaon) required maximum days for sclerotial formation followed by isolate RB-12 (kolhapur).

The significant variations in colony growth and colony morphology were observed on different media. Maximum growth of all isolates (mean colony diameter) was observed on PDA (80.56 mm) followed by Richard's Agar (79.90 mm), while least growth was recorded on Rose Bengal Agar (67.32 mm). In cultural study of isolates on different media, the observations related to colony type, mycelial growth, colony margin, colour of the colony and days required to form sclerotia were recorded. Variation in these characters among the isolates were recorded according to the media.

Pathogenic variability was observed among the isolates in sick soil method as well as in blotter paper technique. Isolate RB-5 (MPKV isolate) was found most virulent followed by RB-4 (Koradgaon, Ahmednagar isolate), which recorded 100 and 93.33 percent disease incidence, respectively. RB-12 (Kolhapur isolate) was found least virulent as disease incidence observed was only 33.33 per cent. Remaining isolates were moderately virulent.

In screening of genotypes against *R. bataticola* the genotypes, viz. Digvijay, Vihar and GNG-469 exhibited moderately resistant reaction whereas four genotypes, viz. PG-03-02-10, Virat, DC-2012-3 and GNG-1581 showed tolerant reaction towards the disease. Remaining 32 genotypes were found moderately susceptible (14) to highly susceptible (18) to the disease.

1. INTRODUCTION

Chickpea (*Cicer arietinum* L.) is the world's third most important food legume after soybean and pea. Out of total production of pulses, chickpea accounts for 20 per cent. India contributes the major share to the global chickpea area (about 65%) and production (65%). In India, it is grown on 9.01 million ha area with a production of 8.83 million tonnes (Anonymous, 2014a). Maharashtra is the third largest chickpea producing state in India after Madhya Pradesh and Andhra Pradesh. In Maharashtra, the crop occupies an area of 18.20 lakh ha with production 16.22 lakh ton and productivity 891 kg/ha. In western Maharashtra region, total area under chickpea was 3.42 lakh ha (Anonymous, 2014b).

In India, Chickpea is primarily grown as *Rabi* (post rainy) season crop on residual soil moisture. It improves physical, chemical and biological properties of soil by fixing atmospheric nitrogen symbiotically. Deep roots of the crop also open up the soil by increasing soil aeration and fit well in various cropping ecosystem and hence has got unique position in rain-fed agriculture.

Chickpea is the best and cheapest source of protein in human diet. It is the rich source of protein(21.1%) carbohydrate (61.5%) and fat (4.5%).Chickpea is mainly used for human consumption as well as animal feed. It is consumed as whole seed, fried, boiled, salted or more generally which is cooked. Fresh green leaves are used as vegetable. Exudation from leaves (locally called *amb*) contain malic acid and oxalic acid which possess medicinal value for bronchitis, diarrhea,

digestive disorder, warts and blood purification. Germinated seed are recommended to cure scurvy disease.

Chickpea productivity often remains low due to susceptibility of the crop to different biotic and abiotic stresses. Among the biotic stresses, diseases, insect pest, nematodes and parasitic weeds account for major losses. The crop is attacked by 172 pathogens which include 67 fungi, 22 viruses, 3 bacteria, 80 nematodes and mycoplasma from all over the world (Nene *et al.*, 1984). Among these, only a few have the potential to devastate the crop. The important soil borne diseases infecting chickpea are *Fusarium* wilt, dry root rot and collar rot.

Dry root rot caused by *Rhizoctonia bataticola* (Taub.) Butler (pycnidial stage: *Macrophomina phaseolina* Tassi Goid) is emerging as serious threat to the chickpea growing regions of India (Masood Ali and Shivkumar, 2001). The pathogen survives in soil on infected host debris. Secondary spread of the disease is through farm implements, irrigation water and rain splash. Recently wilt resistant chickpea cultivars were found succumbing to dry root rot in major growing areas. (Pande *et al.*, 2010). Being soil borne in nature, it is very difficult to control this pathogen. Yield losses due to root rot are more when the conditions are congenial for disease development. Losses to the extent of 70.8 per cent at full podding and 48.9 per cent at pre-harvest stage of the crop have been reported due to the disease (Ahmed and Mohamad, 1986).

R. bataticola is a diverse, omnipresent soil borne fungal pathogen with a wide host range involving more than 500 species in 75 families (Dhingra and Sinclair, 1978). It mainly produces macro-sclerotia (primary source of inoculum) or

pycnidia. It is both soil and seed borne and infects plants mainly from flowering stage to maturity. The disease is reported to be more severe when chickpea crop is exposed to moisture stress conditions (Sharma *et al.*, 2010). Natural population of this pathogen is diverse in terms of pathogenic variability and can be characterized into pathotypes or races by means of biological typing. Previous studies have revealed and reported high levels of pathogenic and genetic variation in *R. bataticola* from different parts of the world and India (Trivedi and Gurha, 2006; Aghakhani and Dubey, 2009; Pande *et al.*, 2010).

Considering the importance of the disease and area under chickpea cultivation in Western Maharashtra, the present investigation was undertaken to work out the variability in *R. bataticola* isolates from chickpea growing areas of Western Maharashtra with following prime objectives-

- 1) To collect root rot infected samples of chickpea from different chickpea growing areas of western Maharashtra.
- 2) To isolate fungal pathogen from root rot samples.
- 3) To prove the pathogenicity of isolates.
- 4) To study the morphological and cultural characteristics of the isolates.
- 5) To test the response of promising chickpea genotypes against dry root rot pathogen.

2. REVIEW OF LITERATURE

Chickpea is one of the important *Rabi* season pulse crop in India. Production of crop is affected as it is infected by many soil borne pathogens. An attempt has been made in this chapter to compile the literature pertaining to dry root rot pathogen in context to the present investigation entitled, “Morphological variability in *Rhizoctonia bataticola* (Taub) Butler isolates causing dry root rot of chickpea in Western Maharashtra”, on following aspects:

- Occurrence and Yield losses
- Collection, Isolation, Pathogenicity
- Symptomatology
- Cultural and morphological variability
- Screening against the pathogen

2.1 Occurrence and yield losses

2.1.1 Occurrence

Narsimhan (1929) reported the association of fungi with wilted gram plants, *viz.* *Rhizoctonia spp.* and *Fusarium spp.* This appears to be the first report on *Rhizoctonia* wilt of gram from India.

Dastur (1935) reported occurrence of two different wilt of gram from Central provinces of India and isolated species of *Rhizoctonia* resembling to *R. bataticola* from wilt infected gram plants.

Agrawal and Sarbhoy (1976) reported the wide spread occurrence of *Rhizoctonia bataticola* (*M. phaseolina*) causing charcoal rot of Soybean in many parts of india. Pathogen was found to infect the crop in almost all stages of crop growth.

Tripathi and Sharma (1983) reported the incidence of *R. bataticola* on chickpea in southern Haryana. It was revealed that incidence was high in late October to mid-November, decreased in December and January, and again increased in February- March.

Pandey and Singh (1990) conducted survey in Allahabad region of India and reported that chickpea was infected by *Fusarium spp.* and *R. bataticola*.

Vishwadhar and Sarbhoy (1991) reported that *R. bataticola* caused seedling rot, root rot, ashey stem blight and charcoal rot in the crops like soybean, mungbean, sunflower, urdbean, etc. in India.

Baw *et al.* (1992) undertaken survey of chickpea diseases in Myanmar and observed that major diseases infecting the crop were dry root rot (*R. bataticola*) and wilt (*F. oxysporum*).

Singh and Sirohi (2003) observed that dry root rot caused by *R. bataticola* was a serious disease in non-irrigated chickpea areas. The incidence of disease was more in Una and Simour district of Himachal Pradesh.

Gurha and Trivedi (2008) reported that *F. oxysporum*. f.sp. *ciceri* and *R. bataticola* caused wilting and dry root rot diseases, respectively in all districts of Himachal Pradesh.

2.1.2 Yield losses

Nene (1985) reported that root rot, collar rot and wilt were the most serious diseases of chickpea which caused severe yield losses upto 30-60 per cent under favorable conditions.

Ahmed and Mohamad (1986) reported about 50- 71 per cent yield losses in chickpea due to *Rhizoctonia* root rot in Bihar state.

Sen and Kajal Kumar (2000) observed that *M. phaseolina* affected a variety of field and vegetable crops and caused charcoal rot which accounted for crop losses to the tune of 10-20 per cent.

Vishwadhar and Chaudhary (2001) noticed that chickpea was susceptible to several fungal diseases. Amongst them, dry root rot was one of the major production constraints that caused 10-20 per cent annual losses.

Sundervadana *et al.* (2012) observed that dry root rot caused heavy damage (up to 60 per cent) to pulse crops compared to other diseases.

2.2 Collection, Isolation and Pathogenicity

Mitra (1931) isolated strains of *Macrophomina phaseolina* from gram and sesame and reported that these were also able to infect potato.

Bhate (1983) isolated *Rhizoctonia bataticola* from roots of green gram and proved its pathogenicity.

Bhatti *et al.* (1987) isolated the pathogens, *viz.* *Fusarium spp.*, *R. solani*, *M. phaseolina* and *Verticilium albo-atrum* from wilted chickpea plants. These pathogens caused 100 per cent wilting in artificial sick soil.

Vishwadhar and Sarbhoy (1991) isolated *R. bataticola* from pods and seeds of diseased soybean plants.

Prameela Devi and Singh (1998) obtained 56 isolates of *M. phaseolina* from roots, stems, pods and seeds of blackgram and greengram samples collected from North,

South, and Central India.

Patel and Anahosur (2000) found association of soil born fungi, *viz.* *Fusarium* spp., *M. phaseolina*, *Sclerotium rolfsii* with wilted chickpea plants. Isolations were made at 10 days interval from 10th day of sowing up to 90th day of crop growth.

Katariya and Gaur (2007) obtained nine isolates of *R. bataticola* from roots of infected chickpea plants and proved their pathogenicity.

Amrutha *et al.* (2014) isolated *R. bataticola* from chickpea and identified the pathogen based on its mycelial and sclerotial characters and pathogenicity test was proved by soil inoculation method on susceptible variety.

Prasad *et al.* (2014) proved pathogenicity of eight isolates of *Rhizoctonia solani* on chickpea variety C-235. It was found that isolate of Bikaner was highly virulent among all the isolate.

2.3 Symptomatology

Nene *et al.* (1978) observed that continuous black discoloration of pith and xylem vessels of roots and basal shoots was one of the main characteristic symptoms of *R. bataticola* infection in chickpea.

Casas and Diaz (1985) described that dry collar and root rot were main symptoms of *M. phaseolina* in chickpea. Affected plants developed dark brown to black roots and dried up without showing any flaccidity.

Haware *et al.* (1990) described typical symptoms of dry root rot in chickpea. They stated that the plant suddenly dried in the field. The leaves and stem of affected plants usually became straw colored. The tap root was dark and

quite brittle in dry soil. The dark black sclerotia were observed on and within host tissue. Further the disease development was influenced by dry soil condition especially at flowering stage.

Beniwal *et al.* (1992) reported sudden drying of the whole plant was the most conspicuous symptom of chickpea dry root rot disease. The tap root of infected plant became dark, showed signs of rotting and was devoid of most of the lateral and finer roots. The dead root was brittle and its tip was easily broken.

Sharma *et al.* (2012) described the symptoms of dry root rot infecting chickpea as sudden drying of whole plant which were scattered in the field. The symptoms include yellowing, drooping of petioles and leaflets only on the tips. Leaves and stem of affected plants showed straw coloured. The tap root turned black, show sign of rooting and was devoid of lateral and fine roots. The dead roots were quit brittle and showed shredding of bark and lateral roots.

2.4 Cultural and morphological variability

Byadgi and Hegde (1985) isolated *R. bataticola* isolates from different host plant and observed variations in the cultural characters, growth rate, morphology of sclerotia, pycnidia and pycnidiospores.

Sobti and Bansal (1988) isolated *R. bataticola* from root, leaf and seed isolates of groundnut and reported variation in cultural characters among these isolate on different media.

Bansal *et al.* (1990) reported cultural variability among six isolates of *R. bataticola* isolated from different

crops.

Monga and Sheo Raj (1994) identified seven isolates of *R. bataticola* from cotton and observed difference in cultural characteristics *i.e.* sclerotial size and number. Two isolates showed differences in pathogenicity on cotton cv DS-1.

Sharma *et al.* (2004) isolated *M. phaseolina* from different crops and observed marked variation in cultural characters among the isolates.

Meena *et al.* (2006) studied variability among the seven isolates of *M. phaseolina* isolated from maize crop from different agro-ecological zones of India. Variation was observed in cultural and pathogenic characters, colour of colony (*viz.* greyish white, blackish grey and dark black). Sclerotia were classified as round and ablong shape. Hyderabad isolate was found more virulent while Coimbatore isolate was least virulent.

Ganeshmoorthi *et al.* (2007) collected five isolates of *M. phaseolina* from different mulberry growing district of Tamilnadu. Among the five isolates, the isolate from Coimbatore took five days to cover entire plate. The isolate produced maximum number of sclerotia on solid and liquid media as compared to other four isolates.

Aghakhani and Dubey (2009) studied diversity among 27 isolates (23 from chickpea and 4 from other host crops) of *Rhizoctonia bataticola* representing 11 different states of India. The isolates showed variability in virulence test. The disease incidence varied from 31.7 to 100 per cent in sick soil method and in blotter paper technique. Disease scale on various cultivars varied between 3-9 grades. Isolate from Uttar

Pradesh exhibited high level of variability among all the isolates.

Amale (2010) studied the 15 isolates of *R. bataticola* infecting chickpea crop from Maharashtra. Variation in colony diameter was recorded. The colony diameter ranged from 61.33 to 90 mm. Sclerotial size ranged 63.20 x 66.40 to 111.3x120.7 μm and sclerotia were classified into round and oval shape.

Kumar and Gaur (2010) studied 14 isolates of *M. phaseolina* from groundnut crop for their cultural and morphological characters, viz. colony colour, pigmentation, mycelial growth, number of sclerotia, virulence and nutritional requirement. The isolate BKN was highly virulent as compared to other isolates.

Manjunatha and Naik (2011) collected thirty isolates of *R. bataticola* from three major chickpea growing regions of Karnataka and found high variability in their cultural and morphological characters. Among these, twelve isolates were fast in growth, another twelve moderate and the remaining six isolates were slow in growth.

Sundravadana *et al.* (2011) reported variability in *R. bataticola* isolates from infected roots, leaf and seed of black gram, greengram, cowpea, soybean and redgram based on virulence. The isolates also showed variability in the colony morphology on different solid media.

Csondes (2012) studied 50 isolates of *M. phaseolina* from Hungary, Spain and Serbia on six different culture media. The fungus exhibited most intensive growth on maize flour agar followed by sabooured-glucose, malt extract, potato

dextrose, Czapek's dox and watery agar medium.

Gupta *et al.* (2012) identified 40 isolates of *R. bataticola* from chickpea crop in Central India. Isolates were highly variable in their cultural, morphological and pathogenic characteristics and were grouped into 24 groups. It was observed that sclerotial size varied from 72.117 x 57.1 to 106.5 μm and number of sclerotia of isolates had positive correlation with virulence. Disease reaction of the *R. bataticola* isolates varied from 6.6 to 9.0 on 1-9 scale.

Rajamohan and Balabaskar (2012) identified 20 isolates of *M. phaseolina* infecting groundnut crop from Cuddalore district of Tamilnadu. The isolates exhibited cultural and pathogenic variability. The isolate MP-11 was having faster mycelium growth, maximum sclerotial size and production and recorded maximum incidence of root rot disease under pot culture condition.

Sharma *et al.* (2012) studied the diversity among 94 isolates from *R. bataticola* obtained from dry root rot infected chickpea plants collected from different agro climatic regions of India. The isolates were found diverse in respect to their different cultural and morphological parameters like colony colour, growth pattern, growth rate, mycelial characters, sclerotial initiation time, sclerotial intensity and morphology of the sclerotia.

Kaur *et al.* (2013) studied variability in 61 isolates of *Macrophomina phaseolina* infecting pigeonpea crop from Varanasi and Mirzapur district. The isolates were categorized on the basis of different morphological characters, *viz.* microsclerotia, texture and colour. The isolates showed

variation in pathogenicity, also.

Verma and Pathe (2013) isolated 22 isolates of *R. bataticola* causing root rot soybean from different places in Madhya Pradesh. The isolates did not differ in type of mycelial growth and potato dextrose agar medium was found favorable for luxuriant growth of all the isolates. However, the isolates revealed variability in size of sclerotia (72.1-99.9×57.1-106.5 µm). Isolates with fluffy growth, small size sclerotia and dark black colony were highly virulent, not positively correlated with virulence. In blotter paper technique 40 to 100 per cent mortality was recorded due to 22 isolates of *R. bataticola*.

Iqbal and Mukhatar (2014) reported variability among the 65 isolates of *M. phaseolina* isolated from mungbean crop. Significant differences were detected among the isolates in their radial growth, sclerotial size as well as pathogenicity. Sixteen isolates were fast growing, 11 as slow growing and remaining medium growing. As regards sclerotial size, 26 isolates had small size and remaining 30 isolates had medium size sclerotia. In the pathogenic variability, 10 isolates were less virulent, 8 highly virulent and remaining moderately virulent. However, no relationship was found between the morphological characters and pathogenicity of the isolates.

2.5 Screening against the pathogen

Agarwal and Sarbhoy (1976) evaluated soybean germplasm for resistance against *M. phaseolina*. Sixty four lines were found moderately resistant and four lines, viz. EC-1077, EC-25166, EC-39069 and EC-39797 resistant in field as well as glass house under artificially inoculated conditions.

Singh and Mehrotra (1982) screened chickpea varieties against *R. bataticola* and reported that the cultivars BG-203, BG-543 and Hare chhole -1 showed resistance to *M. phaseolina* when tested in sick soil under field conditions.

Nene *et al.* (1989) evaluated 225 chickpea genotypes for broad based resistance to wilt and major root rots at 24 locations in 11 countries. The genotypes, *viz.* ICC-2862, ICC-9023, ICC-90321, ICC-10803, ICC-11550 and ICC-11551 were resistant to wilt and root rot.

Reddy *et al.* (1991) screened a large number of chickpea accessions for two consecutive years against dry root rot. Out of the 1883 accessions evaluated, 177 accessions showed less than 20 per cent mortality in the first year of screening and of these 40 accessions showed less than 10 per cent mortality in second year of screening.

Bhatt (1993) evaluated reaction of chickpea cultivar to *R. bataticola* under sick soil condition. The lines, *viz.* BGM-416, BJ-416, ICC-1376 and ICC-11314 were resistance to the pathogen.

Gupta (1995) identified chickpea genotypes with dual resistance against wilt and root rot. Genotype ICCV-89402, ICCV-90201, ICCV-90254, BG-209, H-88-2, GF-88426 and RSG-180 showed less than 10 per cent incidence of both the disease.

Khalid and Ilyas (2000) evaluated 108 chickpea germplasm in green house against dry root rot disease and reported that none of the tested line was highly resistance. Only one line (97112) was resistant and another line (97104) moderately resistant.

Gangawar *et al.* (2002) tested 35 chickpea cultivars against dry root rot in sick plot. None of the cultivar was found immune, while only 10 entries, *viz.* IICC-2644, IICC-10384, IICC-10610, IICC-11224, IICC-11332, ICCL-81002, ICCL-81010, ICC - 12263, ICC-12441 and ICCV- 2 proved to be resistant.

Prajapati *et al.* (2003) evaluated 45 chickpea cultivars for resistance to dry root rot. Among these cultivars H-355, JG-315, Pusa-212, G-543, ICC-32, BG-244, Phule G -5, IICC-10384, IICC-11224, IICC-10630, IICC-11332, ICCV-10, ICC-12441, ICC-12263, ICCL-81010 and Avrodhi showed resistant reaction.

Pande *et al.* (2004) screened chickpea genotypes for their resistance to dry root rot using blotter paper technique and reported that a cultivar (ICCV-2) and breeding line (ICC x 830203-BH-BH-11H) showed resistance to dry root rot, remaining 12 lines were moderately resistant, 19 susceptible and three were highly susceptible.

Mohd. Shaikhul *et al.* (2005) reported that out of 56 chickpea varieties screened for their resistance against *R. bataticola*, only one variety BGD-98 was resistant and three varieties, *viz.* BG-1108, BGD-117, RSG-143 were moderately resistant.

Jayalakshmi *et al.* (2008) screened chickpea genotypes for their resistance to dry root rot in pot culture experiment during *Rabi* season of 2004-2005 and 2005-2006. Out of 12 genotypes tested, four were found resistant, two moderately susceptible, three susceptible and three highly susceptible to dry root rot.

Amale (2010) screened 51 chickpea genotypes against dry root rot pathogen by blotter technique. Genotype Rajas and Virat exhibited resistant reaction, while five genotypes were moderately resistant, three tolerant and remaining were moderately to highly susceptible.

Gupta *et al.* (2012) screened 170 chickpea genotypes against dry root rot by blotter paper technique. Sixty eight genotypes exhibited resistant reaction and 14 moderately resistance reaction against the disease.

3. MATERIALS AND METHODS

The details of the material used and methods followed during the course of present investigation are presented in this chapter.

3.1 Materials

The following material was used during the course of present investigation.

3.1.1 Diseased sample

Chickpea plants showing typical dry root rot symptoms were collected from different district of Western Maharashtra including Pulses Improvement Project, Mahatma Phule Krishi Vidyapeeth, Rahuri (Table 1).

Table 1. Collection of dry root rot samples of chickpea from twelve locations of Western Maharashtra

Location	District	Variety
Anagar	Solapur	Vijay
Piliv	Solapur	Local
Takali-haji	Pune	Digvijay
Koradgaon	Ahmednagar	Local
Pulses Improvement Project, MPKV Rahuri.	Ahmednagar	Vijay
Sonjamb	Nashik	Digvijay
Gavhan	Sangali	Local
Ranjani	Satara	Local
Dhuldeo	Satara	Local
Oil Seed Research station, Jalgaon	Jalgaon	Digvijay
College of Agriculture,Pune	Pune	Vijay
Pishavi	Kolhapur	Jaki

3.1.2 Culture media

The Common laboratory medium *i.e.* Potato Dextrose Agar (PDA) was used for isolation of fungal pathogen responsible for dry root rot of chickpea and pure cultures of isolated organism were maintained on PDA slants for further studies. Different culture media, *viz.* Richard's Agar, Rose Bengal Agar, Czapek's Dox Agar Medium and Potato Dextrose Agar were used to study the cultural and morphological characteristics of *R. bataticola* isolates.

The composition of the media is as follows:

A. Potato Dextrose Agar

Potato (peeled)	200 g
Dextrose	20 g
Agar agar	20 g
Distilled water	1000 ml

B. Richard's Agar

Potassium nitrate	10 g
Potassium dihydrogen phosphate	5 g
Magnesium sulphate	2.5 g
Ferrous chloride	0.02 g
Sucrose	50 g
Agar agar	20 g
Distilled water	1000 ml

C. Czapek's Dox Agar

Sucrose	30 g
Sodium nitrate	2 g
Dipotassium hydrogen phosphate	1 g
Potassium chloride	0.5 g
Magnesium sulphate	0.5 g
Ferrous sulphate	0.01 g

Agar agar	20 g
Distilled water	1000 ml

D. Rose Bengal Agar

Glucose	10 g
Peptone	5 g
Potassium dihydrogen phosphate	1 g
Magnesium sulphate	0.5 g
Streptomycin	30 mg
Agar agar	15 g
Rose Bengal	0.035 g
Distilled water	1000 ml

E. Sand Maize Medium

(for mass multiplication of *R. bataticola* culture)

Sieved sand	800 g
Crushed maize	200 g
Distilled water	100 ml

3.1.3 Seed

The seeds of susceptible chickpea variety BG-212 and promising chickpea genotypes were obtained from the Pulses Improvement Project, MPKV Rahuri.

3.1.4 Glassware

Standard Borosil make glassware, *viz.* Petri plates, test tubes, conical flasks, beaker, etc. were used during the course of research work.

3.1.5 Equipment

Common laboratory equipments used were autoclave, laminar air flow, BOD incubator, refrigerator, research microscope, hot air oven, blender, etc.

3.1.6 Miscellaneous material

Electronic balance, blotter paper, cork borer, earthen pots, forceps, inoculating needles, labels, maize seeds, non-absorbent cotton, plastic buckets, polythene bags, scales, scalpels, spirit lamp, test tube stand, wash bottles, slides, sieve, etc. were used during the course of studies.

3.2 Methods

3.2.1 Isolation

The roots and stem of infected plants were washed in tap water to remove soil adhered to them. The root/ stem were split open and small bits of size 3-4 mm were cut with sterilized blade. These bits were then disinfected with 0.1 per cent aqueous solution of mercuric chloride for two minutes and subsequently washed three times with sterilized water to remove traces of mercuric chloride. Each bit was blot dried and four bits each were placed on the solidified PDA plates. These plates were incubated at $28 \pm 1^{\circ}\text{C}$ for seven days. The well isolated fungal growth free from any contamination was transferred to agar slants by hypal tip method to obtain pure culture. The pure cultures of isolates were maintained on PDA slants for further studies.

3.2.2 Pathogenicity

The pure culture of each isolate was multiplied on sand maize medium for 15 days at room temperature in sterilized flask. The full grown inoculum of each isolate multiplied on sand maize medium was mixed @ 100 g/kg soil. Inoculated soil was incubated for 15 days for growth of fungi in

it. The earthen pots were sterilized with five per cent copper solution and filled with sick (inoculated) soil. Ten seeds of chickpea variety BG-212 were sown in each plot. Three pots were used for each isolate. Seed sown in sterilized (un-inoculated) soil served as control. The observations on symptoms produced on chickpea plants were recorded and per cent disease incidence was worked out by following formula-

$$\text{Percent Disease Incidence (PDI)} = \frac{\text{Number of infected plants}}{\text{Number of plants inoculated}} \times 100$$

3.2.3 Reisolation

Reisolations were made from the roots of artificially infected plants to confirm Koch's postulate. The fungal growth obtained was transferred to PDA slants for comparing with original culture.

3.2.4 Identification of pathogen

The pathogen isolated from diseased samples of chickpea was identified on the basis of morphological characters and symptoms produced on plants under sick soil condition.

3.2.5 Morphological and Cultural studies

Isolates of *R. bataticola* were grown on different media, viz. Richard's Agar, Rose Bengal Agar, Czapek's Dox Agar and Potato Dextrose Agar in order to study their growth characters. The plates were poured with respective media and inoculated with 5 mm diameter disc of each fungal isolates at centre. The plates were incubated at $28 \pm 1^{\circ} \text{C}$ for 8 days. The

observations on mycelial growth, nature of growth, colony colour, time required for sclerotial formation, etc. were recorded.

To study the characters, *viz.* sclerotia size and shape each of the isolates was grown on PDA medium and microscopic observations were recorded.

3.2.6 Variability studies

To know the pathogenic variation among the 12 isolates, dry root rot susceptible chickpea variety BG-212 was evaluated against all isolates of *R. bataticola* by blotter paper technique (Nene *et al.*, 1981).

The seeds were surface sterilized with 2 per cent sodium hypochlorite for two min, rinsed in sterile water for 2-3 min and sown in earthen pots filled up to 2/3 of its volume with sterilized river sand. The inoculum of each of the isolate was prepared on potato dextrose broth (PDB). For this, 100 ml PDB was placed into 250 ml flask and autoclaved. The sterilized PDB was inoculated with each of the isolate's pure culture separately and incubated for seven days at 28 ± 1 °C. After seven days, mycelial mat was removed from the flask and macerated with 100 ml sterilized distilled water (2 mats each) in blender for one minutes. This prepared inoculum was placed in a beaker. Seven days old seedlings of variety BG-212 grown in small earthen pots were uprooted and washed in sterilized distilled water. Later on, the seedlings were held in hand and roots were deeped in inoculum for about 30 seconds. These seedlings were then placed side by side on blotter paper with one fold so that only the cotyledons and roots remain covered. The blotter paper was

moistened adequately but not excessively. The folded blotter papers were kept in trays and trays were placed in an incubator at 35° C.

Observations were recorded after eight days for the extent of root damage as per the scale given by Nene *et al.* (1981) and Mean Disease Score (MDS) and Per cent Disease Intensity (PDI) was calculated by following formula.

$$\text{MDS} = \frac{\text{Sum of scores}}{\text{Total number of plants assessed}}$$

$$\text{PDI} = \frac{\text{MDS}}{\text{Maximum grade}} \times 100$$

Rating scale for blotter paper technique (Nene *et al.*, 1981)

Rating	Symptoms	Infection on roots (%)	Mean disease reaction
1	No infection on roots	-----	Resistant (R)
3	Very few small lesions on roots	10 % or less	Moderately Resistant (MR)
5	Lesions on roots clear ; but small, new roots free from infection	11 -20%	Tolerant (T)
7	Lesions on roots, many new roots generally free from lesions	21-50%	Moderately Susceptible (MS)
9	Roots infected and completely discoloured	51 % or more	Susceptible (S)

The per cent disease incidence exhibited by the 12 isolates on susceptible variety BG-212 in soil inoculation method

(3.2.2) was also used as criteria to determine the pathogenic variability.

3.2.7 Evaluation of chickpea genotypes against *R.bataticola*

The promising chickpea genotypes as well as varieties released by Pulses Improvement Project, Mahatma Phule Krishi Vidhyapeeth, Rahuri were evaluated for their reaction against *R. bataticola* isolate obtained from Pulse Improvement Project, MPKV, Rahuri by the method mentioned in 3.2.6. The reaction of these varieties /genotypes against the pathogen was recorded and the varieties / genotypes were categorized as per the mean disease reaction.

4. EXPERIMENTAL RESULTS

Chickpea (*Cicer arietinum*) is most important pulse crop in Maharashtra as well as India. The crop suffers from various diseases which causes severe reduction in the grain yield. Now a days, dry root rot is emerging as a threat to the crop. Yield losses up to 10-20 per cent in India have been reported due to the disease. (Vishwadhar and Chaudhary, 2001).

The present investigation was carried out with a view to study the variability in *Rhizoctonia bataticola* infecting chickpea crop. The study includes morphological, cultural and pathogenic variability among the isolates of *R. bataticola*.

4.1 Isolation and Identification

The incidence of dry root rot was mostly observed during flowering to podding stage of the crop. Typical symptoms observed under field condition were yellowing, drooping of leaflets and petioles. Leaves and stem of plants were straw coloured. The plants were easily uprooted and tap roots was black, rotted and devoid of any lateral and fine roots (Plate 1). Isolation made from roots of infected chickpea samples yielded the growth of *Rhizoctonia bataticola*. Out Of 12 samples, four samples each also produced the growth of *S. rolfsii* and *Fusarium* spp. (Table 2). The *Rhizoctonia* isolates were purified by hyphal tip method, sub cultured and maintained on PDA slants for further studies as per the title of the thesis. The isolates were referred as RB (Plate 2).

The twelve isolates from roots of infected chickpea plants were identified as *Rhizoctonia bataticola* on the basis of morphological characters, viz. radial hyaline colonies which

Table 2. *R. bataticola* isolates obtained from twelve different locations of Western Maharashtra

Isolate No.	Location	District	Variety	Growth Stage	Fungus Isolated
RB-1	Anagar	Solapur	Vijay	Flowering	<i>R.bataticola</i> <i>Sclerotium</i> spp.
RB-2	Piliv	Solapur	Local	Flowering	<i>R.bataticola</i>
RB-3	Takali-haji	Pune	Digvijay	Pod formation	<i>R.bataticola</i> <i>Fusarium</i> spp.
RB-4	Koradgaon	Ahmednagar	Local	Pod formation	<i>R.bataticola</i> <i>Sclerotium</i> spp.
RB-5	Pulses Improvement Project, MPKV Rahuri.	Ahmednagar	Vijay	Flowering to pod formation	<i>R.bataticola</i> <i>Fusarium</i> spp.
RB-6	Sonjamb	Nashik	Digvijay	Flowering	<i>R.bataticola</i>
RB-7	Gavhan	Sangali	Local	Flowering	<i>R.bataticola</i> <i>Fusarium</i> spp.
RB-8	Ranjani	Satara	Local	Flowering to pod formation	<i>R.bataticola</i> <i>Sclerotium</i> spp.
RB-9	Dhuldeo	Satara	Local	Flowering to pod formation	<i>R.bataticola</i>
RB-10	Oil Seed Research station, Jalgaon	Jalgaon	Digvijay	Pod formation	<i>R.bataticola</i>
RB-11	College of Agriculture, Pune	Pune	Vijay	pod formation	<i>R.bataticola</i> <i>Fusarium</i> spp.
RB-12	Pishavi	Kolhapur	Jaki	Pod formation	<i>R.bataticola</i> <i>Sclerotium</i> spp.

later became carbonaceous brown to black, mycelium septate and dark brown in colour, typical right angled branching of mycelium and production of sclerotia.

4.2 Pathogenicity

The pathogenicity test of *Rhizoctonia* isolates was carried out by soil inoculation method (pot culture) and blotter paper technique on susceptible variety BG-212. In sick soil, typical root rot symptoms were observed within 45-50 days after sowing. The plants turned straw coloured as dry grass, root became black and broken easily. Whereas, in blotter paper technique blackening of roots was clearly visible after eight days of incubation. Thus, all the isolates were found pathogenic under both sets of experiments (Plate 3).

The fungus isolated from these diseased plants (grown in pots) was found identical with the original culture and the morphological characters were confirmed under microscope.

4.3 Variability studies

4.3.1 Morphological variability

4.3.1.1 Variation in colony diameter

Variation in colony diameter of the *R. bataticola* isolates was recorded when grown on PDA medium (Table 3, Plate 4). Accordingly, isolates were grouped into i) fast growing (more than 80 mm) ii) moderately growing (in between 70 to 80 mm) and iii) slow growing (less than 70 mm). Fast growing isolates were RB-1, RB-3, RB-4, RB-6, RB-8, RB-9 and RB-11, wherein average diameter of colony ranged from 81.33 to 90 mm. Moderately growing isolates were RB-2 and RB-7 and the mean colony diameter was 76.20 and 79.33 mm respectively, while isolates RB-10 and RB-12 were observed to be slow growing (60.66 and 62.66 mm, respectively).

4.3.1.2 Variation in colony morphology

A. colony growth

Most of isolates produced cottony and compact mycelium. Isolates RB-1, RB-2, RB-4 and RB-6 produced ring like mycelium, whereas RB-7 and RB-10 produced thread like mycelium. Dense and compact colony was observed in isolates RB-3, RB-5, RB-8 and RB-9; less compact colony was observed in isolates RB-10 and RB-12. The colony margin of all the isolates was circular except isolate RB-8 which had irregular margin (Table 3).

B. Colony colour

The colony colour of the isolates varied from grey to black. It was black in case of isolates RB-4, RB-5, RB-9 and RB-11, whereas other isolates had whitish grey, grey, smoky grey and grey to black colour of colony (Table 3).

4.3.1.3 Days required to form sclerotia

On the basis of days required to form sclerotia isolates were grouped into two groups, *viz.* early sclerotial formation (up to 8 days) and late sclerotial formation (more than 8 days). Isolates RB-1, RB-3, RB-4, RB-5, RB-6 and RB-8 showed early sclerotial formation, while isolates RB-2, RB-7, RB-10, RB-11 and RB-12 showed late sclerotial formation (Table 3).

4.3.1.4 Size and shape of sclerotia

The size and shape of sclerotia are presented in Table 4. The sclerotial size varied among the isolates and maximum sclerotial size (113.52 x 104.06 μm) was recorded in isolate RB-6 while minimum (54.39 x 52.03 μm) was in isolates RB-8. Based on the size of sclerotia isolates were

Table 3. Colony characteristics of *R.bataticola* isolates on PDA medium

Isolate No.	Mean colony diameter (mm)	Colony margin	Colony colour	Days required to form sclerotia	Mycelial character
RB-1	85.16	Circular	Gray	08	Mycelium ring like, less compact
RB-2	79.33	Circular	Gray to black	09	Mycelium dense, ring like
RB-3	81.33	Circular	Whitish gray	07	Mycelium cottony, dense, compact
RB-4	88.16	Circular	Black	07	Mycelium ring like, compact
RB-5	90.00	Circular	Black	06	Mycelium dense, thread like, cottony, compact
RB-6	86.00	Circular	Smoky gray	07	Mycelium cottony, ring like, compact
RB-7	76.20	Circular	Whitish gray to black	09	Mycelium cottony, less compact, thread like
RB-8	82.00	Irregular	Gray to black	07	Mycelium cottony, dense, compact
RB-9	90.00	Circular	Black	08	Mycelium dense, cottony, compact
RB-10	62.66	Circular	Gray to black	12	Mycelium cottony, less compact, thread like
RB-11	85.33	Circular	Black	09	Mycelium slightly cottony, compact, profuse
RB-12	60.66	Circular	Smoky gray	10	Mycelium cottony, less compact
SE ($\pm$)	1.35				
CD at 5 %	3.96				

grouped into i) small size (below 60 μ m) ii) medium size (60 to 100 μ m) and iii) large size (above 100 μ m). Isolates RB-1, RB-4, and RB-8 showed small size of sclerotia, isolates RB-5 and RB-6 large and remaining isolates showed medium size

Table 4. Morphology of sclerotia of different *R.bataticola* isolates on PDA medium

Isolates No.	Shape of sclerotia	Size (Length × Breadth) (μm)
RB-1	round	56.76 × 52.03
RB-2	oval	66.22 × 59.12
RB-3	round	68.58 × 66.22
RB-4	round	60.07 × 58.17
RB-5	round	102.64 × 100.27
RB-6	oval	113.52 × 104.06
RB-7	oval	69.53 × 62.43
RB-8	round	54.39 × 52.03
RB-9	oval	73.31 × 65.27
RB-10	round	85.14 × 83.72
RB-11	oval	70.95 × 63.85
RB-12	round	69.53 × 66.22

sclerotia. The isolates produced either round or oval shaped sclerotia. In case of isolates RB-1, RB-3, RB-4, RB-5, RB-8, RB-10 and RB-12 sclerotia were round shaped and in remaining isolates the shape was oval.

4.3.2 Cultural variability

4.3.2.1 Variability in growth of isolates on different media

A. Richard's Agar

Observations on mean colony diameter, colony morphology and days required to form sclerotia were recorded on and are presented in Table 5.

Mean colony diameter of the isolates ranged from 60.50 mm (RB-12) to 90 mm (RB-3 and RB-5). Profuse colony was observed in isolates RB-1, RB-8 and RB-9. Isolates RB-1, RB-2, RB-4 and RB-6 showed ring like mycelium while it was thread like in isolates RB-5, RB-7 and RB-10.

Table 5. Colony characteristics of *R.bataticola* isolates on Richard's Agar medium

Isolate No.	Mean colony diameter (mm)	Colony margin	Colony colour	Days required to form sclerotia	Mycelial character
RB-1	86.00	Circular	Whitish gray	06	Mycelium ring like, dense, profuse
RB-2	75.33	Circular	Gray to black	08	Mycelium dense, ring like, compact
RB-3	90.00	Circular	Smoky gray	06	Mycelium dense, cottony, compact,
RB-4	82.33	Irregular	Blackish	06	Mycelium ring like, less dense,
RB-5	90.00	Irregular	Gray to black	05	Mycelium less dense thread like
RB-6	87.33	Circular	Gray	06	Mycelium cottony, ring like, less dense
RB-7	75.33	Circular	Gray to black	08	Mycelium dense, less compact, thread like
RB-8	83.33	Circular	Gray to black	07	Mycelium cottony, less dense, profuse
RB-9	81.00	Circular	Black	06	Mycelium cottony, cottony, profuse
RB-10	63.33	Circular	Smoky gray	10	Mycelium cottony, less compact, thread like
RB-11	84.33	Circular	Gray to black	09	Mycelium slightly cottony, dense ,compact,
RB-12	60.50	Circular	Smoky gray	09	Mycelium cottony, compact, dense
SE ($\pm$)	1.24				
CD at 5%	3.63				

On the basis of colony type (mycelial growth) observed, the isolates were grouped as having dense colony and less dense

colony. Isolates RB-2, RB-3, RB-7, RB-11 and RB-12 showed compact and dense colony while it was less dense in isolates RB-4, RB-5, RB-6 and RB-8. According to colony margin isolates were classified into circular and irregular margin groups. All isolates were having circular margin except isolate RB-4 (irregular margin). The isolates exhibited three types of colony colours: whitish grey to smoky grey (RB-1, RB-3, RB-6, RB-10 and RB-12), grey to black (RB-2, RB-5, RB-7 RB-8 and RB-11) and black (RB-4 and RB-9) (Plate 5).

On the basis of days required to form sclerotia, isolates were grouped into early sclerotial formation (5-8 days) and late sclerotial formation (above 8 days). Isolates RB-10, RB-11, and RB-12 showed late sclerotial formation, while early sclerotial formation was observed in rest of the isolates.

B. Czapek's Dox Agar

On Czapek's dox agar medium, minimum colony diameter was recorded with isolate RB-12(60.33 mm), whereas maximum was that of isolate RB-5 (87.33 mm). Isolates RB-3 and RB-9 showed profuse colony growth. In case of mycelium growth, isolates RB-4 and RB-6 showed ring like mycelium, while it was thread like in isolate RB-5. As regards colony type, the isolates RB-3 and RB-7 showed dense colony, whereas remaining less dense colony. Compact growth of colony was observed in isolate RB-2, RB-7, RB-10 and RB-12. Only the isolate RB-3 showed irregular colony margin while all other had circular colony margin (Table 6).

Variation was also recorded in colony colour of the isolates. Isolates RB- 2, RB-6, RB-10 and RB-12 showed whitish grey to smoky grey; isolates RB-1, RB-4, RB-7, RB-8

Table 6. Colony characteristics of *R.bataticola* isolates on Czapek's Dox Agar medium

Isolate No.	Mean colony diameter (mm)	Colony margin	Colony colour	Days required to form sclerotia	Mycelial character
RB-1	75.66	Circular	Gray to black	08	Mycelium less dense ring like,
RB-2	70.66	Circular	Gray	09	Mycelium less dense, compact
RB-3	71.00	Irregular	Smoky gray	08	Mycelium dense, cottony, compact, profuse
RB-4	81.16	circular	Smoky gray to black	08	Mycelium ring like, less dense,
RB-5	87.33	Circular	Black	07	Mycelium less dense, cottony, thread like
RB-6	84.66	Circular	Gray black	07	Mycelium , ring like, less dense
RB-7	76.50	Circular	Gray to black	08	Mycelium dense, less compact, thread like
RB-8	79.66	Circular	Gray to black	08	Mycelium cottony, less dense
RB-9	81.66	Circular	Blackish	09	Mycelium cottony, cottony, profuse
RB-10	62.00	Circular	Whitish gray	12	Mycelium cottony, less compact
RB-11	80.83	Circular	Gray to black	07	Mycelium slightly cottony, dense
RB-12	60.33	Circular	Smoky gray	10	Mycelium cottony, less compact, less dense
SE ($\pm$)	0.94				
CD at 5%	2.76				

and RB-11 grey to black and isolates RB-5 and RB-9 black coloured colonies (Plate 6).

In case of days required to form sclerotia, isolates RB-2, RB-9, RB-10 and RB-12 showed late sclerotial

formation (above 8 days). In remaining isolates early sclerotial formation (up to 8 days) was recorded (Table 6).

C. Rose Bengal Agar

Table 7. Colony characteristics of *R.bataticola* isolates on Rose Bengal medium

Isolate No.	Mean colony diameter (mm)	Colony margin	Colony colour	Days required to form sclerotia	Mycelial character
RB-1	68.66	Circular	Gray to black	09	Mycelium less dense ring like, growth away from centre
RB-2	62.83	Circular	Gray to black	09	Mycelium less dense, less compact
RB-3	74.83	Circular	Black	08	Mycelium less dense, cottony, profuse
RB-4	67.66	Circular	Smoky gray	09	Mycelium ring like, very less mycelium formed
RB-5	78.33	Circular	Black	07	Mycelium less dense, cottony,
RB-6	70.00	Circular	Gray to black	07	Mycelium , ring like, growth away from centre
RB-7	72.83	Circular	Gray black	08	Mycelium dense, compact, thread like
RB-8	70.16	Circular	Whitish gray to black	07	Mycelium less compact, dense
RB-9	69.83	Circular	Blackish	09	Mycelium cottony, compact and dense
RB-10	55.66	Irregular	Whitish gray	12	Very less one sided growth mycelium
RB-11	63.33	Circular	Smoky gray to black	07	Mycelium slightly cottony, less dense
RB-12	53.83	Circular	Smoky gray	10	Not more mycelium formed
SE ($\pm$)	0.67				
CD at 5%	1.95				

The morphological and cultural characters recorded on Rose bengal agar medium are presented Table 7. The range

of mean colony diameter of isolates was 53.83 mm (RB-12) to 78.33 mm (RB-5). Isolate RB-1, RB-4 and RB-6 showed ring like mycelium, while isolate RB-7 had thread like mycelium. The isolates RB-7 and RB-9 exhibited dense colony type, whereas remaining isolates had less dense colony. The colony margin of isolate RB-10 was irregular, while in all remaining isolates it was circular

In case of colony colour, isolates RB-4, RB-7, RB-10, RB-11 and RB-12 showed whitish grey to smoky grey colour, isolates RB-1, RB-2, RB-6 and RB-8 grey to black and in isolates RB-3, RB-5 and RB-9 it was black (Plate 7). Early formation of sclerotia (up to 8 days) was recorded in the isolates RB-3, RB-5, RB-6, RB-8 and RB-11 while remaining isolates showed late formation of sclerotia (above 8 days).

4.3.3 Pathogenic variability

The pathogenic variability among the 12 *R. bataticola* isolates was studied on dry root rot susceptible variety BG-212 by sick soil (pot culture) method as well as blotter paper technique.

From the results presented in Table 9, it is revealed that, all isolates were pathogenic and in sick soil method, the disease incidence ranged between 33.33 per cent and 100 per cent. However, the incidence varied among the isolates and accordingly the isolate were grouped into three categories as i) less virulent (30-60 per cent incidence) ii) moderately virulent (61-90 per cent incidence) iii) highly virulent (above 100 per cent). The isolates RB-6, RB-10 and RB-12 showed least virulent reaction while, isolates, viz. RB-4, RB-5 and RB-11 highly virulent and remaining isolates expressed moderately

Table 9. Pathogenic variability among different isolates of *R. bataticola* (Cv-BG-212)

Isolates No.	Mortality (%) (Sick soil method)	Percent Disease Intensity (%) (Blotter paper technique)
RB-1	76.66	85.55
RB-2	80.00	88.88
RB-3	86.60	93.33
RB-4	93.33	100.00
RB-5	100.00	100.00
RB-6	53.33	68.88
RB-7	86.60	95.55
RB-8	76.66	84.44
RB-9	83.33	94.44
RB-10	46.66	61.11
RB-11	90.00	97.77
RB-12	33.33	54.44
Control	00.00	00.00

virulent reaction. Maximum disease incidence (100%) was recorded by isolate RB-5 whereas minimum (33.33%) by isolate RB-12.

In blotter paper technique, cultivar BG-212 showed susceptible disease reaction to all the isolates. Variable levels of per cent disease intensity were recorded by the isolates indicating the variability in virulence of isolates. Accordingly, the isolates were grouped into two groups as i) less virulent (intensity up to 70 per cent) ii) highly virulent (above 70 per cent). Isolates RB-6, RB-10 and RB-12 showed less virulent reaction while remaining all isolates exhibited highly virulent reaction.

4.3.4 Evaluation of chickpea genotypes against *R. bataticola* using blotter paper technique

The 39 promising chickpea genotypes were

screened against the most virulent isolate (RB-5, MPKV isolate) for their reaction to dry root rot disease. The result presented in Table-10 revealed that none of the genotypes was resistant to the disease. The genotypes, *viz.* Digvijay, Vihar and GNG-469 exhibited moderately resistant reaction whereas four genotypes, *viz.* PG-03-02-10, Virat, DC-2012-3 and GNG-1581 showed tolerant

Table 10. Screening of chickpea genotypes against *R. bataticola* by blotter paper technique

Sr. No.	Genotype	Per cent disease intensity after 8 days	Mean disease reaction
01	PG-12310	22.44	MS
02	PG-12404	66.66	S
03	PG-0611	72.64	S
04	PG-12302	24.32	MS
05	PG-0405	77.78	S
06	PG-018108	22.32	MS
07	PG-12107	78.43	S
08	PG-12110	76.43	S
09	PG-408	72.34	S
10	PG-0302-10	18.22	T
11	PG-06-09-15-2	76.80	S
12	PG-10114	28.64	MS
13	PG-12407	62.42	S
14	Digvijay	9.60	MR
15	Virat	19.82	T
16	L-550	100.00	S
17	Vijay	35.80	MS
18	Vishal	28.20	MS
19	Vihar	9.80	MR
20	Kripa	62.80	S
21	GNG-2171	52.50	S

22	NDG-11-12	29.50	MS
23	H-09-96	28.80	MS
24	BDG-1074	28.00	MS
25	H-09-90	77.80	S
26	PG-01-20	25.70	MS
27	GJG-1001	24.00	MS
28	GL-28295	24.90	MS
29	DCP-92-3	64.50	S
30	GJG-1010	30.00	MS
31	GL-28297	100.00	S
32	GCP-101	100.00	S
33	IPC 2006-126	96.40	S
34	JG-315	100.00	S
35	GNG-469	8.30	MR
36	DC-2012-3	15.40	T
37	GNG-1581	11.63	T
38	GCP-105	25.15	MS
39	BG-212	100.00	S

reaction towards the disease. Out of the remaining 32 genotypes, 14 were moderately susceptible and 18 highly susceptible to the dry root rot disease.

5. DISCUSSION

The present investigation was carried out with a view to study the variability in *Rhizoctonia bataticola* infecting chickpea crop in Western Maharashtra. The study included morphological, cultural and pathogenic variability. The results obtained from investigation are discussed in this chapter.

5.1 Collection of diseased samples and Isolation of pathogen

In present studies, diseased chickpea plants showing typical symptoms of dry root rot were collected from 12 different locations in Western Maharashtra. Isolation made from roots of these infected chickpea samples yielded the growth of *R. bataticola*. The other fungi, viz. *Fusarium* spp. and *sclerotium rolfsii* were also found associated with diseased plants as four samples each produced the growth of these fungi along with *R. bataticola*. The twelve isolates of *R. bataticola* were further studied for variability as per the title of thesis. The identification of isolated fungus was done at Department of Plant Pathology and Agricultural Microbiology, MPKV, Rahuri on basis of morphological characters observed under microscope.

Several workers have reported the pathogen associated with dry root rot of chickpea. Aghakhani and Dubey (2009), Manjunatha and Naik (2011), Gupta *et al.* (2012), Prasad *et al.* (2014) and Amrutha *et al.* (2014) had isolated *R. bataticola* from dry root rot infected chickpea plants.

Similarly, Bhatti *et al.* (1987) and Patel and Anahosur (2000) found association of fungi like *Fusarium*

spp., *Sclerotium rolfsii* and *Verticillium albo-atrum* with wilted chickpea plants.

The results of the present study are in the line with the earlier findings, wherein *R. bataticola* was mainly obtained from diseased samples collected during flowering to podding stage of the crop.

5.2 Pathogenicity

Pathogenicity test of all 12 isolates was carried out by soil inoculation method as well as blotter paper technique. Seeds of susceptible variety BG-212 were sown in pots filled with soil inoculated with the *Rhizoctonia* isolates and observations on symptoms produced on plants were recorded. It was noticed that the symptoms on chickpea plants were identical with those observed under natural conditions. Moreover, the typical dry root rot symptoms were recorded by all the 12 isolates under study and the variety BG-212 showed susceptible reaction to all the isolates. The fungal pathogen re-isolated from roots of infected plants grown in artificially inoculated soil was identical to that one isolated from naturally diseased plants samples.

In blotter paper technique, blackening of roots on chickpea seedlings was clearly visible after eight days of incubation. All the isolate induced blackening of roots and the variety BG-212 was found susceptible to 12 isolates.

Earlier Katariya and Gaur (2007) proved the pathogenicity of *R. bataticola* on chickpea plants by soil inoculation method.

Similar results were also reported by Amrutha *et al.* (2014) *R. bataticola* was isolated from chickpea plants and its

pathogenicity was proved by soil inoculation method on susceptible chickpea variety.

Sharma *et al.* (2012) conducted the pathogenicity test of 50 isolates of *R. bataticola* on susceptible chickpea cultivar BG-212 by blotter paper technique. All the isolates were found pathogenic.

In present studies the pathogenicity of 12 isolates was proved by soil inoculation method and blotter paper technique and all the isolates were found pathogenic to dry root rot susceptible variety (BG-212).

5.3 Morphological variability

The morphological characters such as colony diameter, colony morphology, mycelial growth, colony colour, days required to form sclerotia and size and shape of sclerotia of twelve isolates were studied by growing the pure culture on PDA medium

The colony diameter of the isolates ranged from 60.66 to 90 mm and according to colony diameter, isolates were grouped as fast, moderately and slow growing. The isolates showed variation in mycelial growth type. However the isolates were categorized into dense and less dense groups according to the growth observed on the medium. Mycelium was two types: thread like and cottony. Colour of colony was categorized into whitish grey, smoky grey, grey to black and blackish. According to colony margin isolates were grouped into two groups: circular and irregular. Whereas, on basis of days required to form sclerotia, two groups of isolates were observed early sclerotia forming and late sclerotia forming. The sclerotial size ranged from 52.03×54.39 - 104.06×113.52

μm and according to size of sclerotia, the isolates were classified as small size, medium size and large size. Shape of sclerotia was categorized into round and oval shape.

Earlier variation in cultural and morphological characteristics of *R. bataticola* isolates has been reported by many workers.

Monga and SheoRaj (1994) observed differences in sclerotial size and number among seven isolates of *R. bataticola* isolated from cotton.

Meena *et al.* (2006) reported that *M. phaseolina* isolates from maize crop varied in cultural and pathogenic characters including colour of colony. Colony colour was greyish white, blackish grey and dark black colour. Sclerotia were classified as round and ablong shape.

Amale (2010) reported that colony diameter among isolates of *R. bataticola* varied from 61.33 to 90 mm. Sclerotial size varied between 63.20×66.40 to $111.3 \times 120.7 \mu\text{m}$ and according to shape classified the sclerotia as round and oval.

Manjunatha and Naik (2011) found high variability in cultural and morphological characters among the 30 isolates of *R. bataticola* from major chickpea growing areas. The isolates differed in growth rate, also.

Gupta *et al.* (2012) reported that size of sclerotia of *R. bataticola* isolates from chickpea varied from 72.117×57.1 - $106.5 \mu\text{m}$ and classified them as minute, medium and large size.

Sharma *et al.* (2012) studied *R. bataticola* isolates of chickpea from different agro-climatic regions of India. The isolates were found diverse in respect to their different

cultural and morphological parameters like colony colour, growth pattern, growth rate, mycelial characters, sclerotial initiation time, sclerotial intensity and morphology of the sclerotia.

Variation in cultural and morphological characters of *R. bataticola* isolates from different crop has been reported by many workers such as groundnut (Rajamohan and Balabaskar, 2012), pigeonpea (Kaur *et al.* 2013), soybean (Verma and Pathe, 2013) and mungbean (Iqbal and Mukhatar, 2014)

During present studies, variation was observed among the 12 isolates obtained from different chickpea growing area of Western Maharashtra in respect to colony diameter, colony colour, mycelial growth characters, sclerotial size and shape.

5.3 Cultural variability

Cultural variability of isolates was studied on four different media, *viz.* Richard's agar, Czapek's dox agar, Potato dextrose agar and Rose bengal agar medium. Observations related colony diameter and colony morphology were taken. Mean colony diameter of isolates on PDA (80.56 mm), Richard's agar (79.90 mm), Czapek's dox agar (75.95 mm) and Rose bengal agar (67.32 mm) varied accordingly. However, the isolates showed maximum growth on PDA and it was followed by Richard's agar medium. Poor growth was observed on Rose bengal agar medium. Variation in other characters such as colony form, mycelial type (cottony or thread like), colony growth (dense or less dense) and colony colour (whitish and

smoky grey, grey to black) was recorded among the isolates when grown on different media.

The results of the present investigation are in confirmation with Csondes (2012) who reported maximum growth of *R. bataticola* isolates on potato dextrose agar followed by Czapek's dox agar media.

Verma and Pathe (2013) have also reported maximum growth of *R. bataticola* on PDA.

The present results are also in agreement with Amale (2010) and Sundervadana *et al.* (2011).

5.5 Pathogenic variability

The pathogenic variation in isolates of *R. bataticola* was determined by soil inoculation method (glasshouse conditions) and blotter paper technique. In soil inoculation method, the disease incidence in cultivar BG-212 ranged from 33.33 to 100 per cent. On the basis of disease incidence, the isolates were divided into three groups as highly virulent (more than 90 per cent disease incidence), moderately virulent (60-90 per cent incidence) and least virulent (30-60 per cent incidence). The isolate RB-5 was more virulent as it caused 100 per cent incidence followed by RB-4 which caused 93.33 per cent incidence on susceptible variety BG-212.

Variation was also observed in the virulence of the isolates in blotter paper technique and the disease intensity on BG-212 ranged from 54.44 to 100 per cent. The isolates were divided into less virulent and highly virulent groups.

Earlier, Aghakhani and Dubey (2009) studied diversity among 27 isolates (23 from chickpea and 4 from other host crops) of *R. bataticola* and reported that isolates

showed variability in virulence test. The disease incidence varied from 31.7-100 per cent in sick soil method while in blotter paper technique, the disease response on various cultivars varied between 3-9 grades.

Verma and Pathe (2013) reported 40 to 100 per cent mortality in soybean in blotter paper technique due to 22 isolates of *R. bataticola*.

Gupta *et al.* (2012) reported that disease reaction of the *R. bataticola* isolates varied from 6.6-9.0 on 1-9 scale in blotter paper technique.

Pathogenic variability among the *R. bataticola* isolates has been also reported by Monga and Sheo Raj (1994) in cotton, Kumar and Gaur (2010) in groundnut and Kaur *et al.* (2013) in pigeonpea.

Similarly, during present studies pathogenic variation was observed among the 12 isolates when tested by soil inoculation method and blotter paper technique.

5.6 Screening of chickpea genotypes

The 39 promising chickpea genotypes were screened against the most virulent isolate for their reaction to dry root rot disease. Results indicated that none of the genotypes was resistant. The genotypes Digvijay, Vihar and GNG-469 exhibited moderately resistant reaction, while Virat, PG-03-02-10, DC-2012-3 and GNG-1581 showed tolerant reaction and remaining genotypes showed susceptible reaction to the disease. Blotter paper technique is a quickest method recommended for screening chickpea varieties against *R. bataticola*.

Pande *et al.* (2004) screened chickpea genotypes for their resistance to dry root rot using blotter paper technique and reported that only a cultivar (ICCV-2) and breeding line (ICC x 830203-BH-BH-11H) showed resistance to dry root rot, while 12 lines were moderately resistant, 19 susceptible and three were highly susceptible.

Amle (2010) reported that five chickpea genotypes, *viz.* Digvijay, PG-0517, ICCV-06304, PG-0515 and PG-00381 showed moderately resistant reaction, three genotypes PG-97030, ICCV-06102 and ICCV-03407 tolerant reaction and remaining susceptible reaction when tested against *R. bataticola* by blotter paper technique.

Gupta *et al.* (2012) screened 170 chickpea genotypes against dry root rot by blotter paper technique. Sixty eight genotypes exhibited resistant reaction and 14 moderately resistant reaction.

Screening for disease resistance is a continuous process. The resistant sources identified through screening can play an important role as these can be used as resistant doner in breeding for developing disease resistant varieties. Several workers, *viz.* Khalid and Ilyas (2000), Gangawar *et al.* (2002), Mohd. Shaikhul *et al.* (2005) and Jayalakshmi *et al.* (2008) have earlier screened chickpea cultivars against *R. bataticola* for locating the resistance.

6. SUMMARY AND CONCLUSION

6.1 Summary

Chickpea (*Cicer arietinum*) is one of the most important legume crops grown in India. The country is largest producer of the crop in world. Chickpea grown as rainfed, suffers from several biotic and abiotic factors causing economic losses. Among them, dry root rot is most destructive seed and soil borne disease. *Rhizoctonia bataticola* infecting chickpea was first reported by Narasimhan in 1929 in India and confirmed as a dry root rot by Dastur in 1935. Since then there have been various efforts to combat the disease particularly through resistant varieties, chemical and biological control. However, breeding for disease resistant varieties has not proved satisfactorily for control of *Rhizoctonia* root rot of chickpea mainly because of appearance of races in *Rhizoctonia bataticola*. Variability in *Rhizoctonia* pathogen has been reported by many researchers which may be one of the reasons for breakdown of resistance of chickpea varieties.

Therefore, it was felt necessary to study morphological, cultural and pathogenic variation in dry root rot pathogen infecting chickpea in Western Maharashtra. The infected samples of chickpea were collected from the different locations in Western Maharashtra and causal organism was isolated in pure form.

The pathogenicity was proved by pot culture (sick soil) method as well as blotter paper technique with the susceptible cultivar BG-212. Typical symptoms developed on plants when

they were grown in inoculated soil and incubated in blotter paper with inoculum of different isolates.

Morphological characters of twelve *R. bataticola* isolates were studied on PDA medium. Sclerotia were round or oval shape, size of sclerotia varied between 52.03 x 54.39 to 104.06 x 113.52 μm .

The mean colony diameter significantly differed from isolate to isolate and ranged from 60.66 to 90 mm. The mean colony diameter of isolate RB-5 and RB-9 was maximum, while it was minimum in RB-12.

Variation in colony growth of *R. bataticola* isolates was observed and accordingly, isolates were categorized into dense and less dense group. Out of 12 isolates, four isolates showed dense growth.

On the basis of mycelial colour, the isolates of *R. bataticola* were grouped as whitish grey, smoky grey, grey to black and blackish. However, the intensity of colour varied from isolate to isolate. Out of 12 isolates, four isolates showed whitish grey to smoky grey, four grey to black and remaining blackish colour.

As regards days required to form sclerotia, isolates were grouped into early and late sclerotial formers. Isolates RB-2, RB-7, RB-10, RB-11 and RB-12 showed late sclerotial formation and remaining isolates showed early sclerotial formation.

Cultural variability of isolates was tested on four different media. These four media were Potato dextrose agar,

Richard's agar, Czapek's dox agar and Rose bengal agar. The highest growth of all isolates was recorded on PDA and this was followed by Richard's agar, Czapek's dox agar and Rose bengal agar medium. Variation in other colony characters was also recorded on these four culture media.

Pathogenic variability among isolates was tested in pot culture under glasshouse conditions as well as blotter paper technique with susceptible chickpea cultivar BG-212. The isolate RB-5 was more virulent as it caused 100 per cent mortality followed by RB-4 (99.33 per cent) in pot culture. In blotter paper technique, the isolates RB-4 and RB-5 were found to be more virulent (100 per cent disease intensity) followed by RB-7 (95.55% disease intensity).

Chickpea genotypes were screened against most virulent isolate of *R. bataticola* (MPKV isolate) by blotter paper technique. Genotypes were rated as resistant, moderately resistant, tolerant, moderately susceptible, and highly susceptible according to their disease reaction. Genotypes, *viz.* Digvijay, Vihar and GNG-469 showed moderately resistant reaction, whereas four genotypes exhibited tolerant and remaining 32 genotypes showed moderately susceptible to highly susceptible reaction towards the disease.

6.2 Conclusion

- On the basis of colony growth, it was observed that isolate RB-5 and RB-9 (90 mm) were fast growing and this was followed by RB-4 isolate (88.16 mm).

- Among all the isolates, RB-6 had largest size of sclerotia, *i.e.* $104.6 \times 113.52 \mu\text{m}$ and this was followed by RB-5 ($100.27 \times 102.64 \mu\text{m}$). Round and oval shape of sclerotia were mainly observed.
- Isolate RB-10 showed late sclerotial formation (after 12 days) followed by RB-12 (after 10 days), while early sclerotia were formed (5-8 days) in isolates RB-1, RB-3, RB-4, RB-5, RB-6, RB-8 and RB-9.
- PDA proved to be the best medium for growth of all isolates on which average growth of all isolates was 80.56 mm, followed by Richard's agar on which average growth of isolates was 79.90 mm. Rose bengal agar was least favourable for growth of all isolates wherein average growth of isolates was 67.32 mm.
- All the 12 isolates collected from different districts of Western Maharashtra were found pathogenic on susceptible variety BG-212.
- In sick soil method, it was observed that isolate RB-5 was highly virulent among all the isolates as it caused 100 per cent disease incidence and this was followed by RB-4 recording 93.33 per cent disease incidence.
- In blotter paper technique, it was observed that isolate RB-4, RB-5 and RB-11 were highly virulent followed by isolate RB-9.
- In screening experiment, three genotypes, *viz.* Digvijay, Vihar and GNG-469 showed moderately resistant reaction. Four genotypes, *viz.* Virat, DC-2012-3, GNG-1581 and PG-

03-02-10 exhibited tolerant reaction and other moderately to highly susceptible reaction to dry root rot disease.

- Thus, it is concluded that morphological, cultural and pathogenic variability exists in different isolates of *R. bataticola* from Western Maharashtra.

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***Originals not seen**

Fig.1. Growth of *R. bataticola* isolates on PDA.

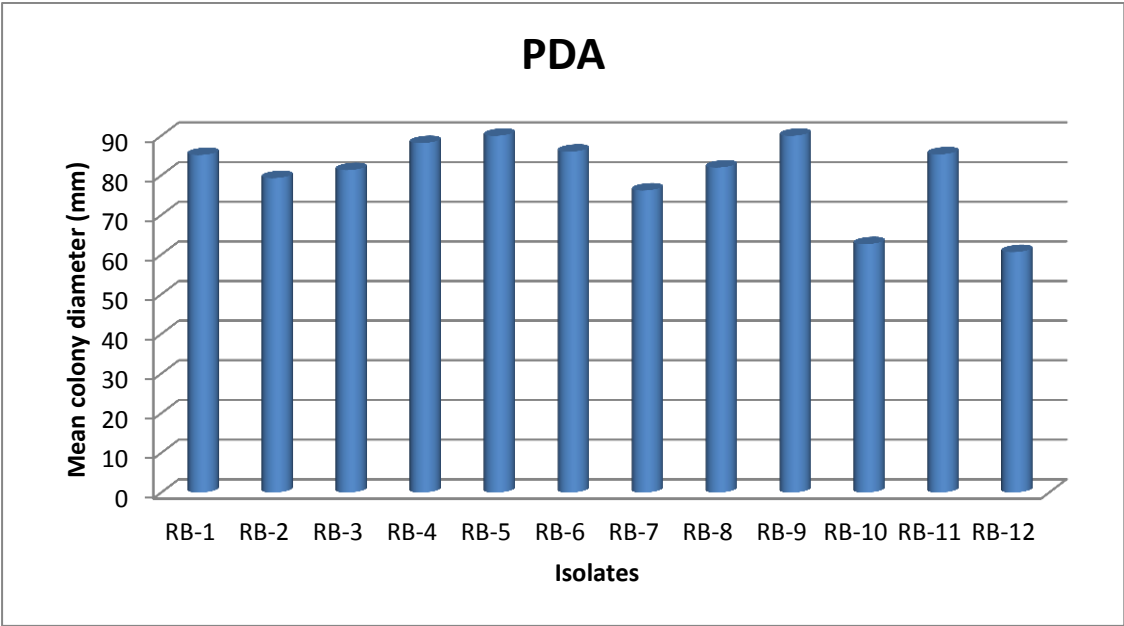


Fig.2. Growth of *R. bataticola* isolates on Richard's Agar Media.

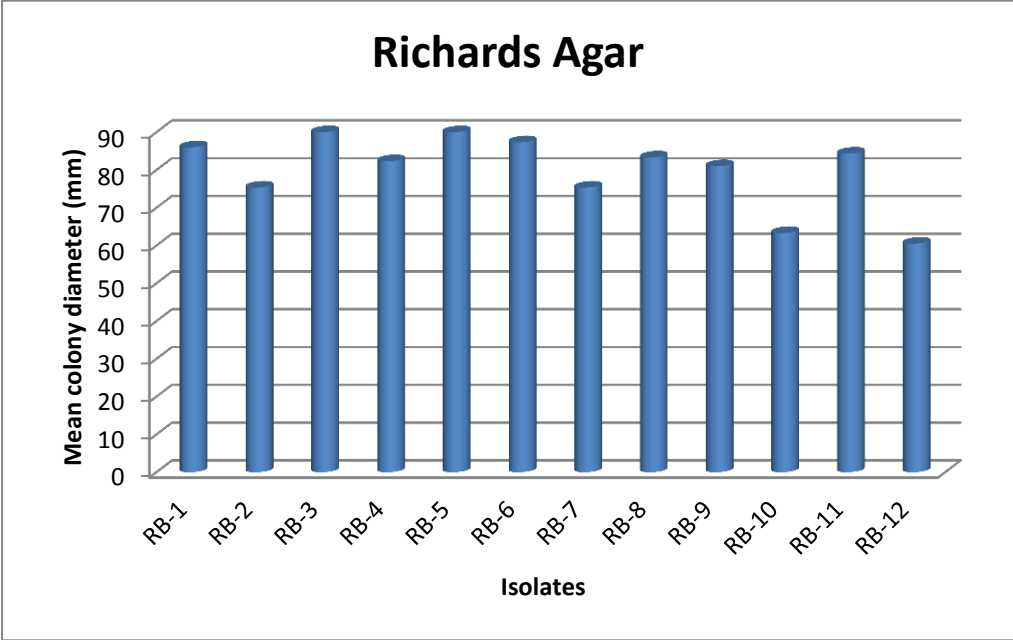


Fig.3. Growth of *R. bataticola* isolates on Czapek's Dox Agar.

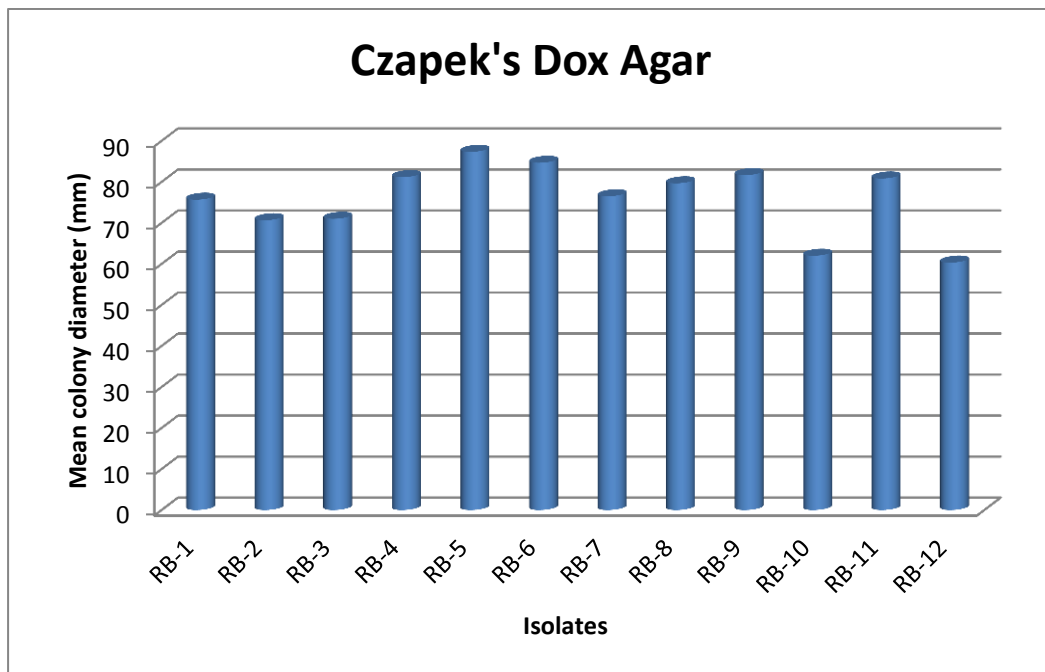


Fig.4. Growth of *R. bataticola* isolates on Rose Bengal Agar.

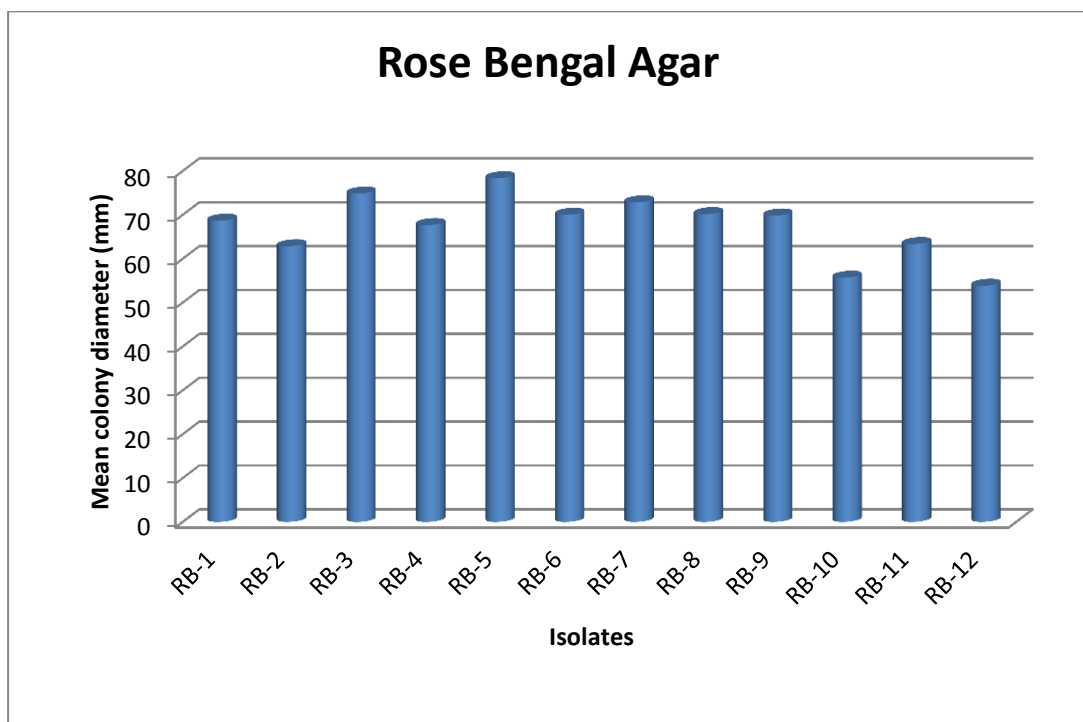


Fig. 5 Comparative growth of isolates of *Rhizoctonia bataticola* on different media.

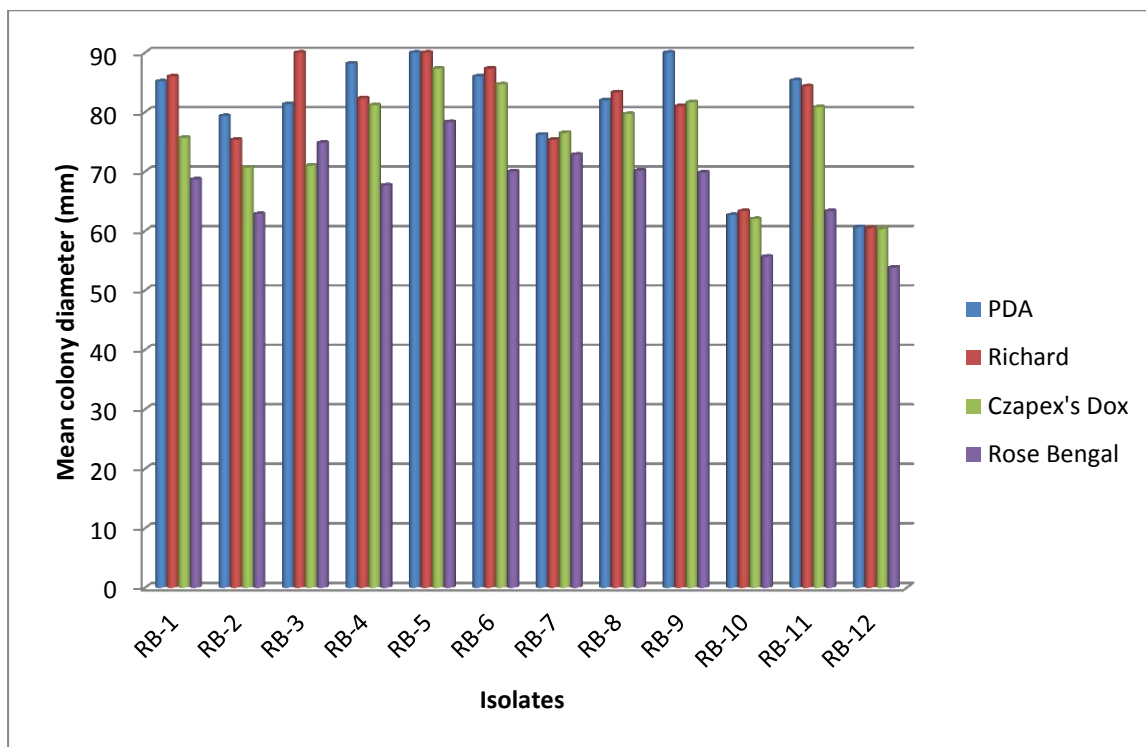
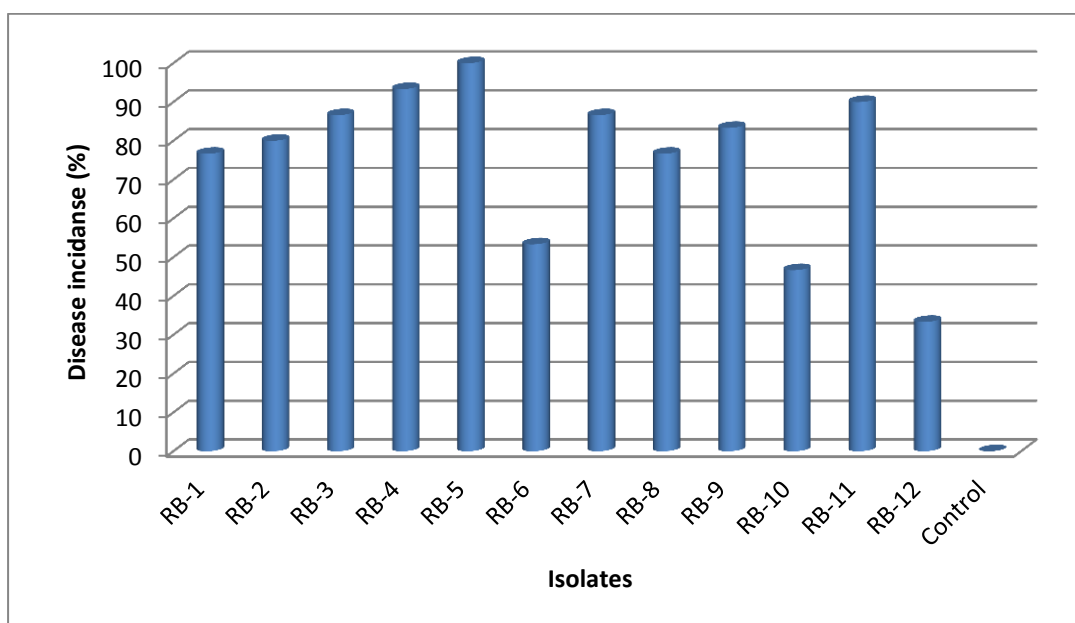


Fig. 6 Disease incidence on susceptible variety BG-212 caused by isolates of *Rhizoctonia bataticola*.



8. VITA

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MASTER OF SCIENCE (AGRICULTURE)
 in
PLANT PATHOLOGY

Title of Thesis : “Morphological variability in *Rhizoctonia bataticola* (Taub.) Butler isolates causing dry root rot of chickpea in Western Maharashtra.”

Major Field : Plant Pathology

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